THE UNIVERSITY OF SYDNEY

Tutorial Class	
Student Name	
SID	

CONFIDENTIAL QUIZ PAPER

This paper should not be removed from the quiz venue.

COMP5339 – Data Engineering
Tutorial Quiz – 1
Individual Quiz (**5%**)

Quiz Writing Time: **30 minutes**
Friday, 22 May, at 07 PM

Instructions to Students

- Please ensure you correctly complete your **tutorial** and **Student ID** details.
- This is a **closed-book** test. **Electronic devices** (including phones, laptops, tablets, etc.) are **not permitted**. Please place all devices in your bag and keep your bag on the floor.
- This tutorial quiz consists of **3 multiple-choice questions** and three short descriptive questions. **All questions must be answered**.
- Write all answers in the spaces provided on this paper.
- Questions carry **different marks**. The mark value for each question is shown at the start of the question. The **total mark** for this quiz is **10**, which contributes to 5% of the total marks.
- For short-answer questions, please write **legibly**. Record your final answers in **ink**. **Do not** use a pencil or **red ink**.

Please submit the completed quiz paper to your tutor **before leaving the room**.

Part I – Multiple Choice Questions

Question 1: This question has three parts. Tick the boxes corresponding to each correct answer. Each question carries one mark.

A. Why is TF-IDF often more informative than raw term frequency alone?

- ☐ It preserves sentence structure
- ☐ It only works for images
- ☐ It gives more weight to words common across all documents
- ☐ **It reduces the weight of terms that are common across documents**

B. A data stream is best described as:

- ☐ A batch file waiting for ETL
- ☐ A fixed-size table stored on disk
- ☐ A normalised relational schema
- ☐ **A potentially unbounded sequence of tuples**

C. Why are message processing guarantees difficult to achieve?

- ☐ A. Because streams are always small
- ☐ **Because producers, brokers, and consumers can fail, causing loss or duplication**
- ☐ Because SQL does not support streaming
- ☐ Because topics cannot be partitioned

Part II – Short Answer Questions

Question 1: Why is feature extraction necessary for text and image data in ML pipelines/workflows? (**2 Marks**)

Feature extraction is necessary because most machine learning algorithms require numerical input, while raw text and images are unstructured. In text, methods such as Bag-of-Words, TF-IDF, and embeddings are used; in images, colour, gradients, texture, shape, or learned neural features can be extracted. This transformation makes the data usable for classification, similarity search, and other ML tasks.

Question 2: What is a session window? Provide one challenge in a streaming pipeline. (**2 Marks**)

A session window is a variable-length window that groups related events based on activity boundaries, such as explicit markers or inactivity gaps. It is useful for modelling sessions such as auctions or user activity periods.

Question 3: Why can in-database machine learning tools such as MADlib be more scalable than the standard approach of loading all data into Python with Pandas for analysis? (**3 Mark**)

MADlib keeps computation close to the data inside the DBMS, reducing data movement and avoiding the need to fit the entire dataset into application memory. It can also exploit database storage, buffering, and in some systems parallel execution. In contrast, exporting data to Python can create memory bottlenecks, data transfer overhead, and limited scalability.

EXTRA SHEET